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Location: St. Louis, Missouri

Generative AI Engineer

PROFESSIONAL SUMMARY

- Results-driven **AI Engineer with 6 years of professional experience in software engineering, including 2+ years focused on AI engineering and Generative AI applications**, across telecommunications, banking, and healthcare domains. Experienced in enterprise application development at **Spectrum, Bank of America, and Optum**, with expertise in Python, FastAPI, RESTful APIs, microservices, backend engineering, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), document intelligence, API integration, and scalable cloud-based AI solutions.Experience developing and supporting **Python-based backend services using FastAPI, RESTful APIs, microservices, SQL, and asynchronous processing**, enabling reliable enterprise integrations, secure data exchange, efficient application performance, and scalable AI-powered business workflows.
- Strong knowledge of **Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and enterprise document intelligence**, including document ingestion, PDF parsing, text preprocessing, semantic chunking, vector embeddings, metadata enrichment, semantic retrieval, and context-aware response generation.
- Familiarity with **AWS Bedrock, Claude LLMs, LangChain, LangGraph, and LlamaIndex**, including LLM integration patterns, prompt engineering, reusable prompt templates, structured JSON outputs, function calling, tool orchestration, and human-in-the-loop validation for enterprise AI applications.
- Understanding of **advanced RAG architectures and retrieval optimization techniques**, including dense retrieval, sparse retrieval, BM25, hybrid search, semantic similarity, metadata filtering, query rewriting, reranking, and grounded response generation to improve retrieval relevance and reduce hallucinations.
- Knowledge of **vector databases and embedding-based retrieval systems**, including FAISS, Pinecone, Amazon OpenSearch, and PostgreSQL/pgvector, with familiarity in vector indexing, similarity search, embedding generation, retrieval evaluation, and enterprise knowledge-base integration.
- Understanding of **Agentic AI architectures and intelligent workflow orchestration**, including AI agents, multi-step task execution, tool calling, conditional routing, stateful workflows, task decomposition, retry mechanisms, error recovery, and human approval checkpoints.
- Familiarity with **LLM evaluation and observability methodologies**, including RAGAS, DeepEval, LangSmith, LLM-as-a-Judge, answer relevance, faithfulness, groundedness, context precision, hallucination analysis, response latency, token consumption, and AI application performance monitoring.
- Experience supporting **AWS cloud services, including Lambda, S3, EC2, Bedrock, and IAM**, along with Docker, Git, CI/CD pipelines, application monitoring, API testing, integration testing, and production troubleshooting for reliable enterprise application delivery.
- Strong understanding of **secure and responsible AI engineering practices**, including OAuth 2.0, JWT authentication, role-based access control (RBAC), PII protection, HIPAA-aligned data security, audit logging, prompt-injection awareness, access controls, and human oversight for trustworthy AI applications.
- Knowledge of **Natural Language Processing (NLP) and intelligent document processing**, including text classification, Named Entity Recognition (NER), keyword extraction, information extraction, document categorization, semantic similarity, and structured data transformation for banking, healthcare, and telecommunications use cases.
- Experienced in collaborating with **cross-functional engineering teams, solution architects, QA engineers, DevOps teams, and business stakeholders** throughout Agile development lifecycles, contributing to requirements analysis, technical design, API integration, testing, debugging, deployment support, and continuous application improvement.

EDUCATION

Master's in Information Technology Management | Webster University, St. Louis, Missouri | May 2024 | GPA: 3.95
Bachelor of Science in Computer Science and Information | Kamala Institute of Technology, Hyderabad, India | GPA: 7.5

PROFESSIONAL SKILLS

Programming Languages	Python, SQL, JavaScript, JSON, Object-Oriented Programming (OOP), Asynchronous Programming, Data Structures, Algorithms, API Development
Backend Development & APIs	FastAPI, RESTful APIs, Microservices Architecture, API Integration, Asynchronous Services, Request Validation, Exception Handling, Pydantic, SQLAlchemy, Authentication, Authorization

Generative AI & LLMs	Generative AI, Large Language Models (LLMs), AWS Bedrock, Claude, Prompt Engineering, Prompt Templates, Structured Outputs, Function Calling, Tool Calling, Context Management, LLM Integration, Hallucination Mitigation, Human-in-the-Loop Validation
RAG & Search	RAG Architecture, Document Ingestion, PDF Parsing, Text Extraction, Recursive Chunking, Semantic Chunking, Metadata Enrichment, Vector Embeddings, Semantic Search, Hybrid Retrieval, Dense Retrieval, BM25, Similarity Search, Reranking, Query Rewriting, Context-Aware Retrieval, Grounded Response Generation
AI Frameworks & Orchestration	LangChain , LangGraph , LlamaIndex , Agent Orchestration Concepts, Stateful Workflows, Tool Routing, Workflow State Management, Human Approval Workflows, Error Recovery
Agentic AI & Automation	AI Agent Architecture, Tool-Calling Agents, Task Decomposition, Conditional Routing, Agent Planning Concepts, Multi-Step Workflows, Function Execution, Retry Mechanisms, Tool Security, Model Context Protocol (MCP) Concepts
Vector Databases & Search	FAISS, Pinecone, Amazon OpenSearch, PostgreSQL/pgvector, Vector Indexing, Embedding Similarity, Metadata Filtering, Hybrid Search, Retrieval Optimization
Natural Language Processing (NLP)	Text Preprocessing, Tokenization, Text Classification Concepts, Named Entity Recognition (NER) Concepts, Keyword Extraction, Semantic Similarity, Document Classification, Information Extraction, Document Intelligence
Machine Learning & Deep Learning	Machine Learning Fundamentals, Model Evaluation Concepts, Embedding Models, Transformer Architecture Concepts, PyTorch, Hugging Face Transformers , Fine-Tuning Concepts, LoRA/QLoRA
LLM Evaluation & Optimization	RAGAS, DeepEval, LangSmith, LLM-as-a-Judge, Retrieval Accuracy, Answer Relevance, Context Precision, Faithfulness, Groundedness, Hallucination Analysis, Latency Optimization, Token Usage Monitoring, Cost Evaluation
Cloud Platforms & AI Services	AWS Lambda, Amazon S3, Amazon EC2, Amazon Bedrock, AWS IAM, AWS CloudWatch Concepts, Cloud AI Deployment Patterns, AWS SageMaker, Azure OpenAI, Google Vertex AI
DevOps & Deployment	Docker, Git, CI/CD Pipelines, Containerization, Cloud Deployment, Application Configuration, Release Management, Integration Testing, Production Support, Application Monitoring
Additional tools	CrewAI, MCP Servers, AWS SageMaker, Azure OpenAI, Google Vertex AI, MLflow, PyTorch, Hugging Face Transformers, LoRA, QLoRA, Kubernetes, Terraform, OpenTelemetry

PROFESSIONAL EXPERIENCE

Client: Spectrum | St. Louis, Missouri | Aug 2024 – Present
Role: AI Engineer
Responsibilities:

- Developed and enhanced scalable **Generative AI applications and enterprise backend services** using Python, FastAPI, RESTful APIs, and microservices architecture, integrating AI-powered capabilities with existing enterprise applications while emphasizing performance, reliability, security, and maintainability.
- Contributed to the design and evaluation of **Retrieval-Augmented Generation (RAG) pipelines** for enterprise document intelligence, including document ingestion, PDF parsing, text preprocessing, intelligent chunking, metadata extraction, embedding generation, semantic retrieval, and context-aware response generation.
- Evaluated **Large Language Model (LLM) integrations using AWS Bedrock**, including Claude-based language models, prompt engineering, reusable prompt templates, structured JSON outputs, function calling, and context management for enterprise knowledge assistants and intelligent automation workflows.
- Explored **LangChain and LangGraph orchestration patterns** for integrating LLMs with retrieval components, enterprise APIs, external tools, and stateful workflows, focusing on modular architecture, controlled execution, and human-in-the-loop

validation.

- Assessed **vector embedding and semantic search approaches** for enterprise knowledge retrieval, including similarity search, metadata filtering, retrieval relevance, and vector database integration patterns involving Amazon OpenSearch, FAISS, Pinecone, and PostgreSQL with pgvector.
- Evaluated **advanced RAG optimization techniques**, including recursive text chunking, semantic chunking, hybrid search, BM25 keyword retrieval, dense vector retrieval, query rewriting, and reranking to improve document retrieval quality and reduce irrelevant context.
- Investigated **agentic AI architecture patterns**, including tool calling, task decomposition, workflow orchestration, conditional routing, execution-state management, and human approval checkpoints for controlled enterprise automation.
- Developed and supported **Python-based API integration services** with request validation, exception handling, authentication, authorization, structured logging, and standardized response formats to ensure reliable communication between enterprise systems.
- Evaluated **LLM response quality and RAG performance** through retrieval accuracy, answer relevance, groundedness, hallucination analysis, response latency, token consumption, and operational reliability.
- Explored evaluation approaches using **RAGAS, DeepEval, LangSmith, and LLM-as-a-Judge** methodologies to assess response faithfulness, context precision, retrieval effectiveness, and prompt performance.
- Supported secure AI integration practices involving **AWS IAM, OAuth 2.0, JWT authentication, role-based access control, sensitive-data handling, and audit logging**, emphasizing enterprise security and responsible AI principles.
- Worked with **AWS services including Amazon Bedrock, Lambda, S3, and EC2** for AI application integration, cloud-based processing, service connectivity, and deployment-related activities.
- Supported **Docker-based application packaging and CI/CD deployment workflows**, collaborating with engineering and DevOps teams to improve application consistency, release reliability, and environment configuration.
- Investigated **AI application observability and monitoring practices**, including request tracing, structured application logs, API error monitoring, model response latency, token utilization, and cloud infrastructure metrics.
- Collaborated with solution architects, AI engineers, backend developers, QA teams, and business stakeholders to analyze requirements, evaluate GenAI use cases, troubleshoot integration issues, and support enterprise AI application delivery.

Client: Bank of America | Hyderabad, India | Oct 2021 – Jun 2022
Role: Software Engineer
Responsibilities:

- Developed, maintained, and supported **enterprise banking backend applications and RESTful API integrations**, ensuring reliable transaction processing, secure communication, consistent data exchange, and seamless integration between internal financial systems.
- Worked on **backend service development, SQL-based data processing, and enterprise application integrations**, implementing business logic, data validation, exception handling, and error recovery mechanisms for critical banking workflows.
- Supported **Python-based automation evaluation and data processing initiatives** to identify opportunities for reducing repetitive tasks, improving operational efficiency, and simplifying financial application workflows.
- Contributed to the assessment of **intelligent document processing and Natural Language Processing (NLP)** approaches for extracting information from banking documents, identifying relevant text, and supporting automated document classification.
- Explored **financial document intelligence concepts**, including text extraction, document preprocessing, information classification, entity identification, and structured data extraction from financial documents.
- Evaluated potential applications of **machine learning and NLP techniques**, including text classification, keyword extraction, Named Entity Recognition (NER), and document categorization for financial information processing.
- Participated in discussions regarding **AI-assisted automation and enterprise knowledge retrieval**, assessing opportunities to improve document accessibility, operational workflows, and information processing.
- Developed and maintained **SQL queries, data validation logic, and database integration workflows** to support accurate financial data processing, application reporting, and downstream system integrations.
- Implemented and supported **REST API validation, request-response processing, exception handling, and integration testing**, ensuring reliable communication across distributed banking applications.
- Assisted in troubleshooting **application performance issues, database errors, API failures, and transaction-processing discrepancies**, performing root-cause analysis and supporting corrective actions.
- Supported **secure application development practices**, including authentication, authorization, access restrictions, sensitive financial information handling, and audit logging to maintain enterprise banking security requirements.
- Participated in **unit testing, integration testing, code reviews, defect resolution, Git-based source control, and CI/CD release activities**, improving application stability and deployment reliability.
- Collaborated with business analysts, solution architects, QA engineers, security teams, and DevOps professionals to translate

banking requirements into reliable backend application functionality.
- Contributed to technical documentation, application maintenance, production issue investigation, and continuous improvement initiatives supporting enterprise banking systems.

Client: Optum | Hyderabad, India | Jan 2019 – Sep 2021
Role: Software Engineer
Responsibilities:
- Supported the development, enhancement, and integration of **enterprise healthcare applications, backend services, and RESTful APIs**, enabling reliable information exchange between healthcare systems while maintaining application stability, scalability, and data integrity.
- Assisted in implementing **SQL-based healthcare data processing and transformation workflows**, validating incoming data, maintaining consistent data formats, and supporting accurate information exchange between integrated applications.
- Developed and supported **data validation and automated quality-check processes**, identifying missing values, inconsistent records, processing errors, and data discrepancies to improve healthcare information reliability.
- Participated in **healthcare API integration and backend service maintenance**, supporting secure communication, request validation, exception handling, and reliable application-to-application data exchange.
- Explored potential applications of **Natural Language Processing (NLP) and intelligent document processing** for extracting and organizing information from clinical documents, healthcare records, and other unstructured medical text.
- Evaluated **Generative AI and Retrieval-Augmented Generation (RAG) concepts** for healthcare knowledge retrieval, including document ingestion, text preprocessing, semantic search, vector embeddings, and context-aware information retrieval.
- Investigated **clinical document intelligence use cases**, including automated document classification, medical terminology extraction, structured information identification, and improved accessibility of healthcare information.
- Explored **Python-based automation opportunities** for healthcare data validation, document preprocessing, API integration, and repetitive data-processing activities.
- Assisted with **secure healthcare data integration**, emphasizing HIPAA-aligned information handling, authentication, authorization, role-based access control, and audit logging across enterprise applications.
- Supported **healthcare interoperability and data exchange requirements**, collaborating with engineering teams to ensure consistent information transfer, accurate data mappings, and reliable integration between healthcare systems.
- Participated in **REST API testing, integration testing, SQL data verification, application debugging, and defect resolution**, identifying technical issues and supporting improvements in application performance and data accuracy.
- Worked with developers, QA engineers, business analysts, security teams, and healthcare stakeholders to understand functional requirements, investigate data inconsistencies, and maintain dependable healthcare application workflows.
- Contributed to **application documentation, testing activities, issue tracking, and production support**, helping improve operational stability, maintainability, and reliability of enterprise healthcare systems.
- Participated in discussions regarding **AI-assisted healthcare automation**, evaluating potential benefits of NLP, document intelligence, semantic retrieval, and knowledge-based information access while considering healthcare data privacy and security requirements.

CERTIFICATIONS

- AWS Certified Machine Learning Engineer – Associate (MLA-C01) | Amazon Web Services
- Claude Certified Architect – Foundations | Anthropic